

Quadratic functions - Set 1

TMUA practice questions by topic. Specification MM1.3, M4.5, M4.11-M4.14. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 The equation $x^2 + (k - 2)x + (k + 1) = 0$ has two distinct real roots. Find the complete set of possible values of k .
- A** $0 < k < 8$
B $-2 < k < 4$
C $k < -2$ or $k > 4$
D $k < 0$ or $k > 4$
E $k < 0$ or $k > 8$
F all real k
- 2 By completing the square, find the minimum value of $2x^2 - 12x + 23$.
- A** -13
B 5
C 23
D 7
E -5
F 3
- 3 The quadratic $y = ax^2 + bx + c$ has its vertex at $(2, -3)$ and passes through $(0, 5)$. Find $a + b + c$.
- A** -1
B 7
C 1
D 5
E -3
F 3

- 4 For what value of c does the line $y = 2x + c$ touch the parabola $y = x^2 - 4x + 7$?
- A 7
B 2
C -2
D -1
E 0
F 5
- 5 How many real solutions does the equation $x^4 - 5x^2 + 4 = 0$ have?
- A 2
B 4
C 5
D 0
E 1
F 3
- 6 The equation $kx^2 + 4x + k = 0$ has no real roots. Find the complete set of values of k .
- A $0 < k < 2$
B $k > 2$
C $k < -2$ or $k > 2$
D $-2 < k < 2$
E $k < -2$
F $-2 < k < 0$ or $0 < k < 2$
- 7 The quadratic $x^2 + px + q$ has minimum value -4 and one of its roots is 0. Which of the following could be the quadratic?
- A $x^2 - 6x + 8$
B $x^2 - 2x - 3$
C $x^2 + 2x - 8$
D $x^2 - 4x$
E $x^2 + 6x + 5$

- 8 The curve $y = x^2 - 2ax + a^2 + a - 6$ has its vertex on the x -axis. Find the value of a .
- A 0
B 6
C -6
D 3
E 2
F -2
- 9 The roots of $2x^2 + bx + 18 = 0$ are equal. Find the sum of the possible values of b .
- A 36
B 6
C 0
D 12
E 24
F -12
- 10 Which of the following quadratics has its vertex in the second quadrant (where $x < 0$ and $y > 0$)?
- A $y = -x^2 - 2x - 3$
B $y = -x^2 + 2x + 1$
C $y = x^2 - 4x + 5$
D $y = x^2 + 6x + 10$
E $y = x^2 + 4x + 1$
- 11 The graph of $y = x^2 + bx + c$ crosses the x -axis at $(1, 0)$ and $(5, 0)$. What is the y -coordinate of the vertex?
- A -9
B 5
C -4
D -3
E 4
F -5

12 Given that $x^2 + y^2 = 10$ and $x + y = 4$, find the value of xy .

A 4

B 2

C 3

D 6

E -3

F 5